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Emerging frontiers in two-dimensional sliding ferroelectrics



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The emergence of sliding ferroelectricity in 2D van der Waals materials has opened a new paradigm for designing nanoscale nonvolatile memories, where polarization switching is governed by interlayer sliding rather than conventional ion displacement. This perspective highlights materials, strategies, switching dynamics, and device applications of 2D sliding ferroelectric materials. Finally, the critical challenges and future research directions are outlined, aiming to guide the rational design of next-generation sliding ferroelectric materials.

Sliding ferroelectrics

Ferroelectric materials have attracted enduring scientific interest due to their unique physical properties arising from spontaneously aligned electric dipoles that can be collectively switched by external electric fields^{1–3}. This remarkable feature, combined with their nonlinear dielectric response and piezoelectric effects, has enabled widespread applications in nonvolatile memory devices, sensors, acoustic wave filters, and optoelectronic systems^{4,5}. However, conventional bulk ferroelectric materials face significant challenges in meeting the demands of modern electronic device development. Their inherent thickness limitations, mechanical rigidity, and the requirement for relatively high switching fields present fundamental obstacles for applications requiring miniaturization, high-density integration, and mechanical flexibility.

The emergence of two-dimensional (2D) materials has opened new possibilities for ultrathin ferroelectric systems. While these atomically thin materials offer several advantages including exceptional mechanical flexibility and reduced switching fields, their practical implementation faces multiple fundamental challenges. First, the stabilization of ferroelectric order in reduced dimensions is severely constrained by the critical size effect, where depolarization fields become increasingly prominent as material thickness decreases. Second, enhanced surface effects in 2D systems lead to significant interface charge trapping and screening effects. Third, substrate-induced strain and interfacial interactions can substantially modify or even suppress intrinsic ferroelectric properties. Although theoretical studies have predicted promising candidates with unconventional ferroelectric mechanisms, such as WO_2Cl_2 where competing soft modes induce non-collinear ferroelectricity, experimental realizations remain scarce⁶. On the

experiment front, the above mentioned factors have resulted in a remarkably limited family of confirmed 2D ferroelectric materials, with only a few systems such as $\alpha\text{-In}_2\text{Se}_3$ ^{7,8}, CuInP_2S_6 ^{9–11}, $\text{CuInP}_2\text{Se}_6$ ¹², SnTe ¹³, NbOI_2 ¹⁴, and $\text{Bi}_2\text{O}_2\text{Se}$ ¹⁵, demonstrating robust ferroelectricity at 2D limit.

Sliding ferroelectricity in 2D van der Waals (vdW) materials has provided groundbreaking solutions to these long-standing challenges. Distinct from conventional ionic-type ferroelectrics, this novel mechanism originates from interlayer charge transfer in non-centrosymmetric stacking configurations. The concept was first proposed in 2017 by Wu's group¹⁶, they proposed a revolutionary concept through first-principles calculations that even when monolayers possess centrosymmetric structures (lacking intrinsic ferroelectricity), carefully engineered bilayer or multilayer stacking could break spatial inversion symmetry to generate robust out-of-plane (OOP) polarization.

This discovery has fundamentally expanded the scope of ferroelectric materials^{17–20}. Subsequent experimental verifications in various systems, including bilayer WTe_2 ^{21–23}, $\beta\text{-InSe}$ ²⁴, boron nitride (BN)^{25,26}, transition metal dichalcogenides (TMDs, e.g., WSe_2 , MoSe_2 , WS_2 , MoS_2)²⁷, 1T'- ReS_2 ²⁸, $\gamma\text{-InSe}$ ²⁹, $\epsilon\text{-InSe}$ ³⁰, and multilayer hexagonal boron nitride (hBN)³¹ (Fig. 1), have firmly established sliding ferroelectricity as a universal phenomenon in vdW materials. The universality of this phenomenon stems from its geometric origin—any 2D system where sliding modifies interlayer charge asymmetry can, in principle, exhibit sliding ferroelectricity. This fundamental insight has expanded the design space for 2D ferroelectrics beyond the constraints of conventional ionic displacement mechanisms and offers several unprecedented advantages: (i) it effectively circumvents the pinning effects caused by defect charges, significantly

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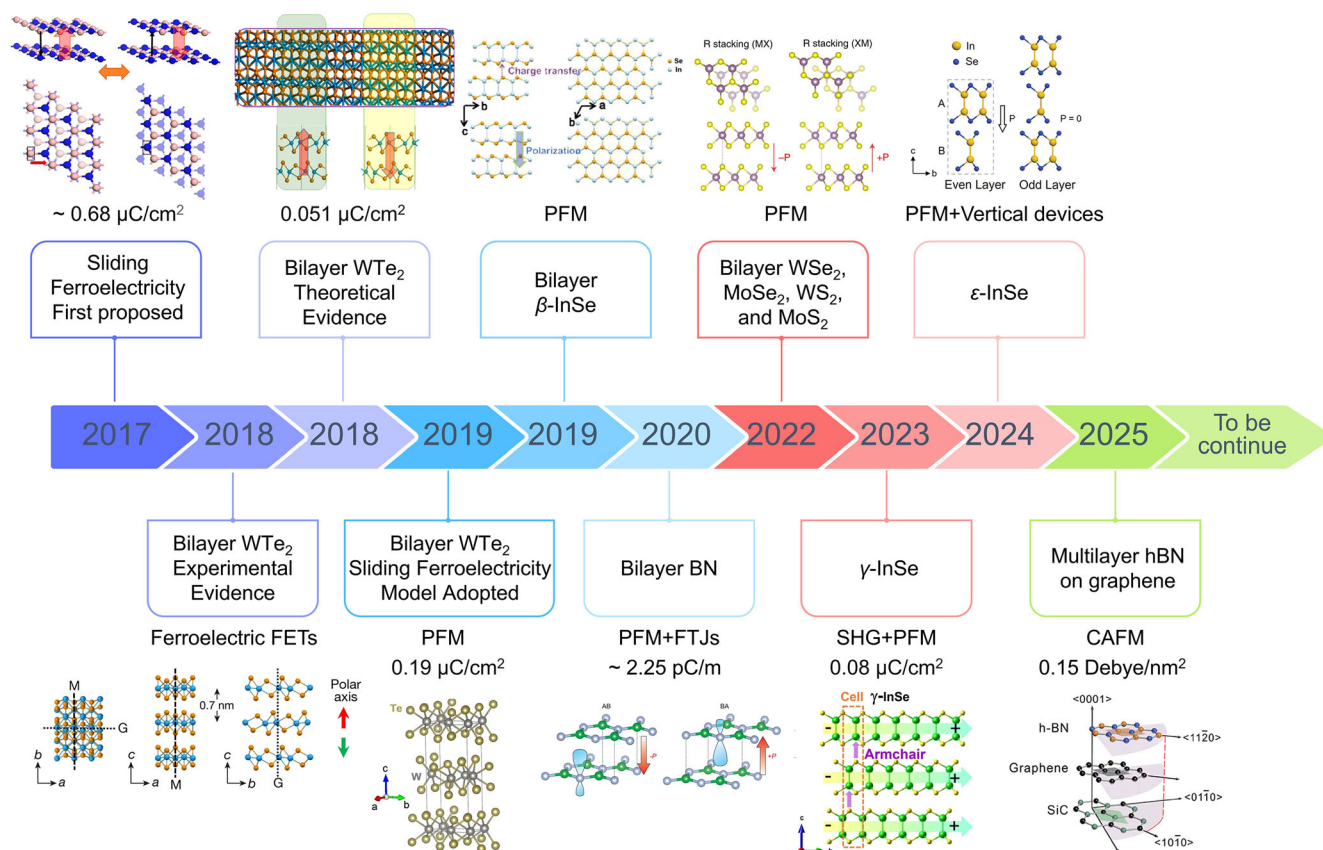


Fig. 1 | The milestones during the development of sliding ferroelectricity.

enhancing endurance and reliability^{32,33}; (ii) the polarization state depends solely on interlayer stacking configuration, enabling novel multistate memory and computing applications; and (iii) the polarization can be modulated through diverse external stimuli including electric fields, mechanical stress, and strains, providing exceptional tunability.

The development of sliding ferroelectricity has opened a new era in ferroelectric research³⁴. In this Perspective article, we highlight representative examples of sliding ferroelectric systems and discuss the fundamental mechanisms behind symmetry-breaking-induced polarization in 2D vdW materials. Rather than providing an exhaustive literature review—as several excellent reviews already cover various aspects of 2D ferroelectrics^{5,19,35,36}—we aim to offer a focused discussion on: (i) sliding ferroelectric material systems, including experimentally confirmed and theoretically predicted sliding ferroelectrics; (ii) strategies for inducing sliding ferroelectricity; (iii) the underlying physics of interlayer sliding-mediated polarization switching; (iv) emerging device applications; and (v) pressing challenges in this rapidly evolving field.

Sliding ferroelectric material systems

Experimentally confirmed sliding ferroelectrics

Inorganic crystals. Graphene, a highly symmetric 2D material, is intrinsically non-ferroelectric in its monolayer and bilayer forms due to its centrosymmetric structure, but theoretical studies proposed by Wu et al. in 2023 have revealed that carefully designed stacking sequences in multilayer graphene ($N \geq 3$) can exhibit unconventional sliding ferroelectricity (Fig. 2a)³⁷. Unlike traditional sliding ferroelectrics where polarization arises from adjacent-layer symmetry breaking, graphene's ferroelectricity originates from cross-layer charge asymmetry, as demonstrated by first-principles calculations showing that specific stackings like ABAC stacking can transform into ABCB by simultaneous translation of two bottom layers by a C-C bond length.

The polarization switching in graphene occurs through distinct multistep pathways involving nonpolar intermediate states (e.g., ABAC $\rightarrow$ ABAB $\rightarrow$ CBAB or ABAC $\rightarrow$ CBAC $\rightarrow$ CBAB), with the asymmetric charge distribution between initial and final states enabling reversible polarization (Fig. 2b). This emergent ferroelectric behavior is highly sensitive to the precise stacking configuration, where only certain carefully designed stacking sequences (e.g., ABAC versus ABCA) can generate the necessary charge asymmetry to support switchable polarization. They further confirmed similar ferroelectric behavior in five-layer graphene systems (Fig. 2c). While scientifically fascinating, these stringent requirements for precise layer control pose significant challenges for practical implementation, highlighting both the potential and limitations of graphene-based sliding ferroelectrics.

As a prototypical vdW dielectric, hBN exhibits unique advantages for realizing sliding ferroelectricity due to its versatile stacking configurations. In its natural AA' stacking configuration, hBN maintains centrosymmetry through antiparallel layer alignment, resulting in a nonpolar state. However, artificial modification of the stacking order to AA or AB/BA configurations breaks inversion symmetry and generates switchable OOP polarization.

A seminal study by Vizner Stern et al. in 2021 first revealed room-temperature ferroelectricity in twisted hBN bilayers using Kelvin probe force microscopy (KPFM), where alternating AB and BA stacking domains showed clear 100 mV surface potential contrast corresponding to opposite polarization states (Fig. 3a)³⁸. This was further corroborated by Yasuda et al. through transport measurements in graphene/hBN heterostructures, where they observed well-defined ferroelectric hysteresis loops with distinct switching characteristics (Fig. 3b)²⁵.

In 2024, Wan et al. demonstrated that strain engineering could also controllably modify ferroelectric domain patterns in hBN (Fig. 3c), creating stable triangular domains through atomic reconstruction effect³⁸. Their

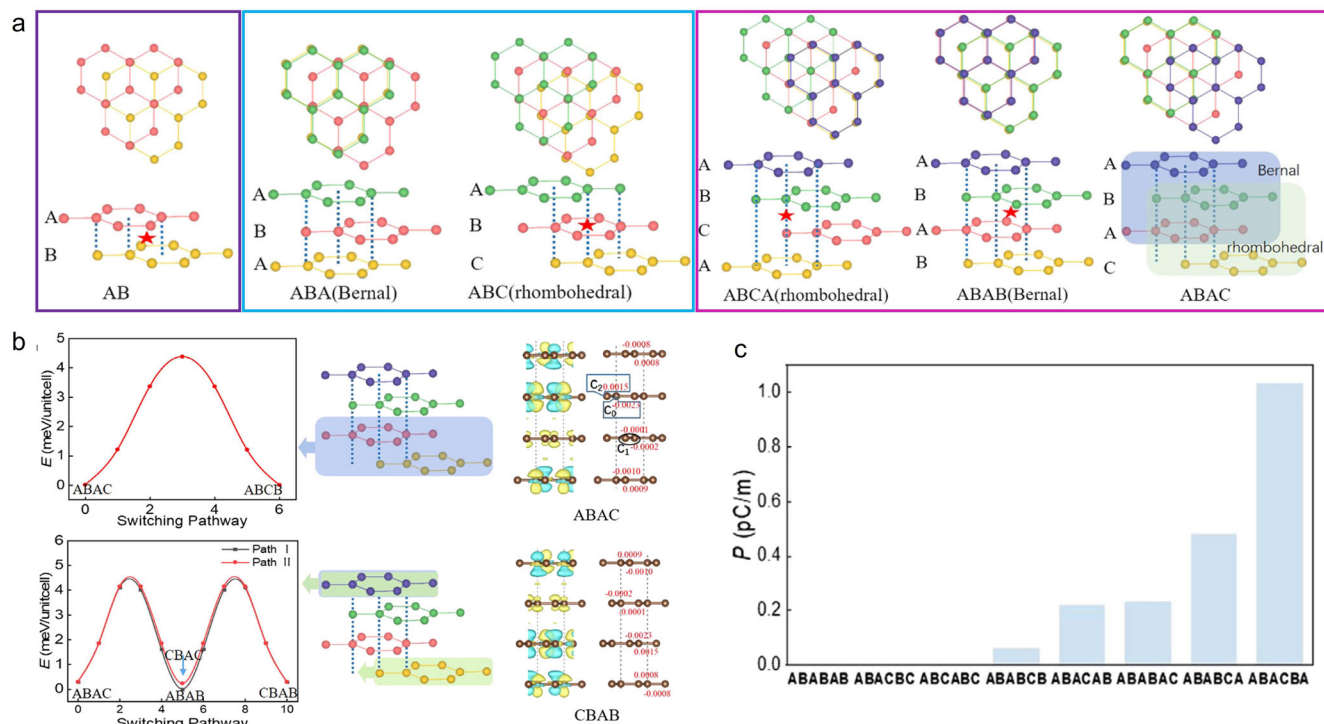


Fig. 2 | Sliding ferroelectricity of graphene. **a** Distinct stacking configurations in bilayer, trilayer, and quadrilayer graphene. **b** Polarization switching pathways between two polarized states in quadrilayer graphene. **c** Stacking configurations and

corresponding theoretical ferroelectric polarization values for pentalayer graphene. Reproduced with permission³⁷. Copyright 2023, American Physical Society.

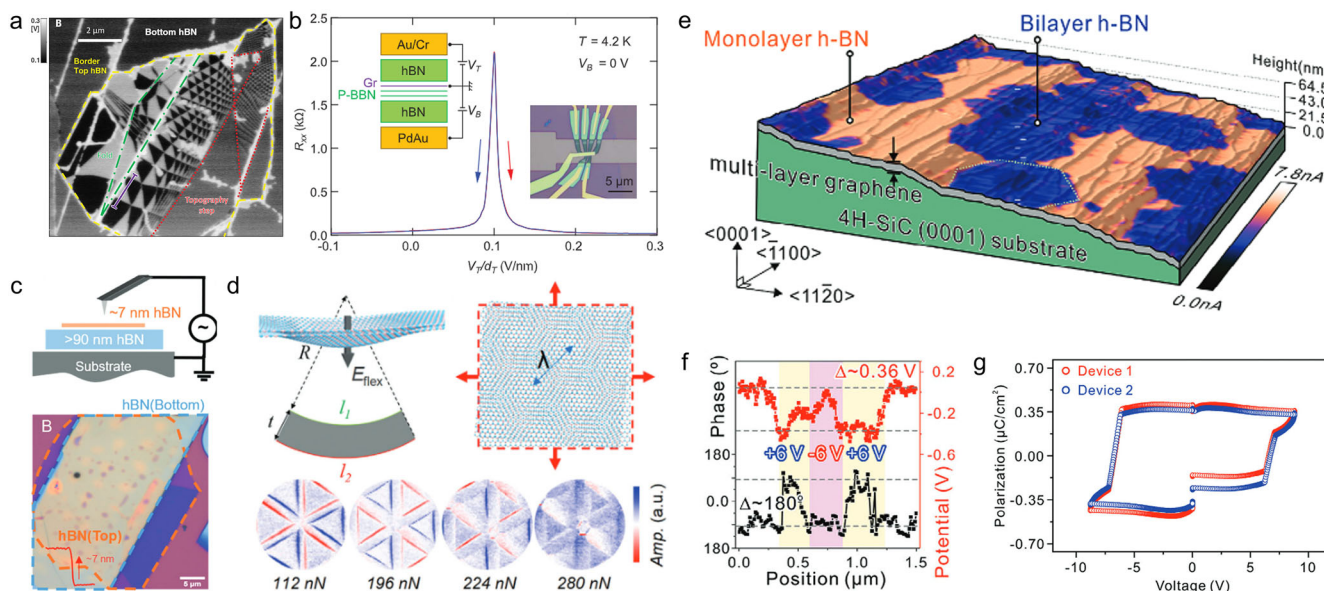


Fig. 3 | Sliding ferroelectricity of hBN. **a** Oppositely polarized domains (black/white regions) corresponding to AB and BA stacking configurations in twisted bilayer hBN. Reproduced with permission²⁶. Copyright 2021, American Association for the Advancement of Science. **b** Ferroelectric hysteresis loop measured in a twisted bilayer hBN-based device. Reproduced with permission²⁵. Copyright 2021, American Association for the Advancement of Science. **c** Schematic structure and optical micrograph of twisted multilayer hBN. **d** Schematic of strain-induced deformation in layered materials under tip pressure, showing moiré

patterns from lattice mismatch and pressure-dependent evolution of OOP ferroelectric domains. Reproduced with permission³⁸. Copyright 2024, John Wiley & Sons, Ltd. **e** Epitaxial growth of hBN on graphene/4H-SiC substrates, showing AFM height and current mapping images. **f** Nested ferroelectric domains ('box-in-box' pattern) in trilayer hBN/graphene heterostructures, with corresponding PFM phase and surface potential variations. **g** Positive-up-negative-down (PUND) measured polarization-electric field hysteresis loops from two distinct five-layer hBN devices. Reproduced with permission³¹. Copyright 2025, John Wiley & Sons, Ltd.

piezoresponse force microscopy (PFM) studies showed these domains could be reversibly switched using local probe tips (Fig. 3d).

The practical viability of hBN-based ferroelectric devices was convincingly demonstrated by Wong et al. through nitrogen plasma-assisted

molecular beam epitaxy growth of hBN/graphene heterostructures (Fig. 3e)³¹. The study revealed intrinsic polarization characteristics induced by moiré superlattices at the hBN/graphene heterointerface, while demonstrating switchable OOP polarization arising from interlayer sliding

in multilayer hBN systems. PFM measurements exhibited 180° phase reversal, and KPFM detected a surface potential difference of 300 mV following polarization switching (Fig. 3f). Their Au/Cr/hBN/graphene devices exhibited robust ferroelectric switching with a coercive field of ± 1.5 V and remnant polarization of $0.375 \mu\text{C}/\text{cm}^2$ (Fig. 3g), maintaining stability for over one week. In 2025, Lin et al. developed a novel sandwiched hBN/rhombic-stacked graphene/hBN heterostructures³⁹, yielding several key breakthroughs: rhombic stacking simultaneously breaks both inversion and mirror symmetries to generate net OOP polarization. Niu et al. discovered that Bernal bilayer graphene moiré superlattice encapsulated within 30° -twisted hBN bilayers could generate an unexpectedly large polarization, producing area charge densities up to 10^{13}cm^{-2} —far exceeding the capacity predicted by moiré band theory⁴⁰.

TMDs represent the most versatile material platform for sliding ferroelectricity, offering distinct advantages over graphene and hBN. In general, TMDs combine following unique advantages: (i) intrinsic noncentrosymmetric crystal phases (e.g., 1 T, 3 R), (ii) widely tunable electronic structures (metallic to insulating bandgaps), and (iii) strong spin-orbit coupling for coupled ferroelectric-spintronic effects. These attributes enable unprecedented design freedom in sliding ferroelectric systems.

To be specific, the polymorphic nature of TMDs allows sliding ferroelectricity to emerge in multi phases, while multilayer stacks exhibit enhanced polarization through interlayer sliding or twist engineering. For instance, 3R-phase MoS_2 bilayers demonstrate switchable polarization *via* $\text{AB} \leftrightarrow \text{BA}$ stacking transitions, a mechanism absent in graphene/hBN systems. Moreover, TMDs' bandgap versatility permits seamless integration with electronic devices, unlike insulating hBN or zero-gap graphene. Recent work has further revealed strain-tunable⁴¹ and light-matter coupled⁴² sliding ferroelectricity in TMDs or their heterostructures⁴³, opening avenues for multifunctional devices. With these advantages, TMDs not only address fundamental challenges in 2D ferroelectrics but also bridge the gap between scientific discovery and practical applications in memory, logic, and optoelectronic technologies⁴⁴.

Organic-inorganic hybrid crystals. Recent advances in sliding ferroelectricity have been primarily limited to indirect characterization techniques due to the inherent challenges of weak polarization and insufficient electrical insulation in most 2D systems. A breakthrough study by Miao et al. addresses this limitation by employing an amphidynamic vdWs crystal, (15-crown-5) Cd_3Cl_6 (CCC), which combines inorganic layers with organic molecular rotators⁴⁵. Unlike conventional 2D ferroelectrics, this hybrid system exhibits a wide bandgap, enabling unambiguous polarization hysteresis measurements at macroscopic scales. Structural and spectroscopic analyses confirm that the ferroelectric response arises from two synergistic mechanisms: (i) dipole formation within individual layers due to rotator freezing, and (ii) interlayer sliding-induced polarization. First-principles calculations further reveal that the net polarization is highly tunable through the interplay between molecular rotation and layer displacement. This work not only provides the first direct evidence of sliding ferroelectricity but also establishes a design paradigm for functional hybrid materials with coupled dynamic and polar properties.

Theoretically predicted sliding ferroelectrics

Beyond above-mentioned typical 2D vdW material systems, recent computational study identifies HgX_2 ($\text{X} = \text{Br}, \text{I}$) as a promising new class of 2D sliding ferroelectrics, expanding the materials palette for slidetrionics applications⁴⁶. The first-principles calculations demonstrate that multilayer HgX_2 adopts stable stacking configurations exhibiting reversible OOP ferroelectricity at room temperature, with HgI_2 showing particularly strong polarization ($0.16 \mu\text{C}/\text{cm}^2$) attributable to pronounced interfacial charge rearrangement during interlayer sliding. Beyond conventional ferroelectric switching, these materials offer unique value through coupled ferroelectric-spintronic functionality—the sliding-controllable Rashba effect in HgI_2 enables electric field manipulation of spin textures, suggesting immediate

potential for 2D spintronic devices. The strong interlayer coupling in HgX_2 distinguishes it from other sliding ferroelectrics like twisted bilayers, potentially offering more robust polarization at reduced layer thicknesses. In addition, some strongly-bonded 2D sliding ferroelectric materials have also occurred in recent years, such as quasi-one-dimensional sliding ferroelectric materials NbI_4 ⁴⁷, kagome- B_2X_3 ($\text{X} = \text{S}, \text{Se}, \text{Te}$) bilayers⁴⁸, and others.

Besides, recent theoretical studies have significantly expanded the family of sliding ferroelectrics, revealing diverse mechanisms and material platforms beyond conventional systems. ZrI_2 bilayers emerge as a prototypical example, exhibiting both in-plane and OOP polarizations due to interlayer charge redistribution⁴⁹. Notably, the β - ZrI_2 phase possesses a sizable OOP polarization ($0.39 \mu\text{C}/\text{cm}^2$) and negative longitudinal piezoelectricity, distinguishing it from other sliding ferroelectrics like CuInP_2S_6 ⁵⁰. Furthermore, ferroelastic-ferroelectric multiferroicity is predicted in ZrI_2 , where 120° ferroelastic switching couples with polarization reversal, enabling six-state logic memory⁴⁹. Charge redistribution in ultrathin β - ZrI_2 also facilitates stable charged domain walls with quasi-2D electron gas, enhancing electromechanical responses⁵¹.

Beyond transition metal halides, pentagonal vdWs heterostructures offer a distinct route to in-plane sliding ferroelectricity. The penta- $\text{PdSe}_2/\text{PtSe}_2$ bilayer exhibits an ultralow switching barrier (1.91 meV/atom) and high polarization ($\pm 17.11 \times 10^{-10} \text{C m}^{-1}$), alongside auxetic behavior (negative Poisson's ratio) for strain-coupled switching⁵². Trilayer NiN_2 ⁵³ further demonstrates the versatility of stacking-engineered ferroelectrics, though detailed mechanisms await exploration. These studies collectively establish design principles for sliding ferroelectrics, emphasizing symmetry breaking *via* stacking, interlayer charge transfer, and strain-polarization coupling, while expanding the material space toward functional heterostructures and multiferroics.

To facilitate systematic comparison of sliding ferroelectric materials, Table 1 summarizes key structural and functional parameters across representative material systems. The table organizes materials by crystal phase and stacking configuration, while listing critical metrics including polarization magnitude, symmetry-breaking mechanisms, and experimental characterization methods. This comprehensive compilation highlights structure-property relationships and identifies gaps in current material exploration.

Engineering Strategies for Breaking Inversion Symmetry

Noncentrosymmetric Crystal Phases

Taking the most extensively studied MoS_2 as an example, its 3 R, 1 T, 1 T, 1 T⁺, and 1 T[−] phases all belong to non-centrosymmetric structural configurations, among which the 3 R and 1 T[−] phases have been successively demonstrated to exhibit sliding ferroelectricity^{54–56}.

In 2024, Yang et al. successfully created switchable ferroelectric domains in 3R- MoS_2 epilayers by inducing a spontaneous shear transformation that generates mobile screw dislocations⁵⁴. These dislocations enable collective polarization switching at remarkably low fields (0.036 V/nm). The resulting ferroelectric transistors exhibit excellent non-volatile memory characteristics, including a 7 V memory window at 10 V operation, retention $>10^4$ seconds, and endurance exceeding 10^4 cycles. In another work, Jiang et al. demonstrates the experimental realization of room-temperature sliding ferroelectricity in chemical vapor deposition (CVD)-grown 3R- MoS_2 bilayers⁵⁵, representing a significant advance toward wafer-scale integration.

Lipatov et al. provide the experimental confirmation of room-temperature OOP ferroelectricity in 2D 1 T[−]-phase MoS_2 ⁵⁶. Through innovative scanning probe manipulation, the authors demonstrate mechanically induced polarization switching in this distorted trigonal phase ($P3m1$ symmetry), creating stable bi-domain states with distinct piezoelectric, second harmonic generation (SHG), surface potential and conductive properties. In another work, through meticulous synthesis of high-purity 1 T[−]- MoS_2 crystals, HuangFu et al. provide conclusive evidence of its noncentrosymmetric structure via atomic-resolution STEM and SHG measurements, while demonstrating reversible polarization switching

Table 1 | Comparative analysis of sliding ferroelectric materials, including phase, stacking type, number of layers, space/point group, polarization intensity, strategies for breaking inversion symmetry, characterization techniques used

Material	Phase	Stacking	Number of layers	Space/point group	Polarization intensity	Strategy	Characterization technique	Ref.
hBN		parallel	2		$P_{2D} = 2.25 (0.37) \times 10^{-12} \text{ C m}^{-1}$	noncentrosymmetric stacking	vertical PFM	25
hBN		parallel	2		$P_z/A = 0.33 \text{ Debye/nm}^2$	noncentrosymmetric stacking	KPFM	26
hBN		Bernal-type	multi		3.01 Debye/nm^2	noncentrosymmetric stacking	KPFM, PUND	31
graphene		Bernal and rhombohedral	>3		0.32 pC/m	noncentrosymmetric stacking		37
hBN			7/90 nm			twist	PFM	38
hBN/graphene		rhombohedral	3		$1.76 \mu\text{C/cm}^2$	heterostructure	device	39
graphene		Bernal	2		5 pC/m		FTJ	40
MoS ₂	2H		2		$d_{33} = 37.54 \text{ pm/V}$	strain	PFM, FTJ, C-AFM	41
MoS ₂	3 R	rhombohedral	2	$P3m1$	$0.8\text{--}1.5 \text{ pC/m}$	photoexcitation		42
MoS ₂ /WS ₂		3R-like and 2H-like	2	$3m$	$d_{33} = 1.95\text{--}2.09 \text{ pm/V}$	heterostructure	PFM	43
MoS ₂	3 R	rhombohedral	2	$C3v$		stacking	device	44
CCC				$P2_1$	$0.3\text{--}0.4 \mu\text{C/cm}^2$	hybrid crystal	PFM	45
Hgl ₂				$Cmc2_1$	$0.16 \mu\text{C/cm}^2$		theoretical	46
Nbl ₄				$Cmc2_1$	$0.11 \mu\text{C/cm}^2$		theoretical	47
kagome-B ₂ X ₃ (X = S, Se, Te)			2	$P\bar{6}2m, P1, P321, Cm, C2$			theoretical	48
Zrl ₂			2	$Pmn2_1$	$0.39 \mu\text{C/cm}^2$		theoretical	49
β -Zrl ₂				C_{1h}			theoretical	51

through both PFM manipulation and FET electrical gating, establishing robust ferroelectric functionality up to a transition temperature of 350 K⁵⁷.

Beyond MoS₂, several other materials with non-centrosymmetric phases also exhibit prominent sliding ferroelectric properties. For example, T-phase MoTe₂ and WTe₂ demonstrate robust sliding ferroelectricity through interlayer charge transfer⁵⁸. In addition, the In₂Se₃ system presents particularly rich polymorphism⁷, with its β , β' , ϵ and γ phases all showing distinct sliding ferroelectric behaviors arising from different symmetry-breaking mechanisms^{59–62}.

In 2022, Zhao et al. achieved centimeter-scale single-crystalline β -, β' -, and α -phase In₂Se₃ thin films through a short-distance CVD method combined with phase transition principles (Fig. 4a)⁷. Inspired by in situ TEM observations showing β -In₂Se's thermal conversion to β' -In₂Se₃, the team introduced β -InSe into precursors to obtain large-area β' -In₂Se₃ films, which subsequently transformed into α -In₂Se₃ via stress relaxation—representing a phase transition from antiferroelectric to ferroelectric ordering (Fig. 4b). Theoretical calculations identified the Se/In ratio as the critical factor governing these phase transformations (Fig. 4c). Field-effect transistors (FETs) fabricated from all three polymorphs demonstrated high carrier mobility (Fig. 4d).

In 2024, Liu et al. achieved precise phase control in In₂Se₃ by utilizing γ -InSe as both precursor and seed material under temperature-regulated conditions (Fig. 4e), successfully mapping the thermal evolution pathways among 3 R α -, β -, and 2H α -In₂Se₃ polymorphs⁶¹. Their work demonstrated the extreme temperature sensitivity of In₂Se₃ formation and phase transitions (Fig. 4f), where both reactive Se vapor concentration and In₂Se₃ growth rates exhibited positive temperature dependence, with β -In₂Se₃ identified as a critical intermediate phase during these transformations. Parallel breakthroughs in understanding phase-dependent ferroelectric behavior came from Che et al.'s atomic-scale investigations, which revealed fundamental differences between 3 R and 2H polytypes through combined STEM and DFT studies⁶². The 2H phase was found to preferentially form in-plane ferroelectric domain walls (Fig. 4g), while the 3 R phase favored OOP

oriented domain walls (Fig. 4h), with these structural distinctions extending to their phase transition mechanisms—the 3 R transition proceeding via relatively simple intralayer atomic sliding, whereas the 2H transition required more complex bond dissociation and reformation processes.

Asymmetric Stackings

In addition to the selection of noncentrosymmetric crystalline phases, the stacking order of layered materials serves as another critical factor determining the degree of symmetry breaking. Recent theoretical advances have established universal design rules for bilayer stacking ferroelectricity (BSF), demonstrating that even two centrosymmetric monolayers can exhibit emergent ferroelectricity through controlled rotation and translation⁶³. In multilayer systems, specific stacking configurations can fundamentally alter the superposition of interlayer electric dipole moments, thereby governing both the orientation and magnitude of ferroelectric polarization. Notably, the BSF theory reveals that polarization directions in bilayers may radically differ from their monolayer counterparts, with out-of-plane and in-plane polarizations becoming interlocked. The resulting polarization is highly sensitive to atomic-scale stacking faults, with even sub-Ångström interlayer displacements capable of switching polarization states.

In 2024, Han et al. demonstrated a substrate-guided CVD growth approach, as illustrated in Fig. 5a for synthesizing SnSe₂ with precisely controlled stacking sequences⁶⁴. Through meticulous interlayer symmetry engineering, they achieved the synthesis of high-order polytypes including 6 R, 12 R, 18 R, and 18 C stacking configurations (Fig. 5b). Notably, clear PFM signals confirming ferroelectric behavior were observed in the AB'-stacked polymorphs, as shown in Fig. 5c, highlighting the critical role of specific interlayer registries in inducing switchable polarization.

However, the CVD epitaxial growth of vdW ferroelectric materials typically struggles to form energetically unfavorable stacking configurations. Consequently, breaking crystallographic inversion symmetry requires

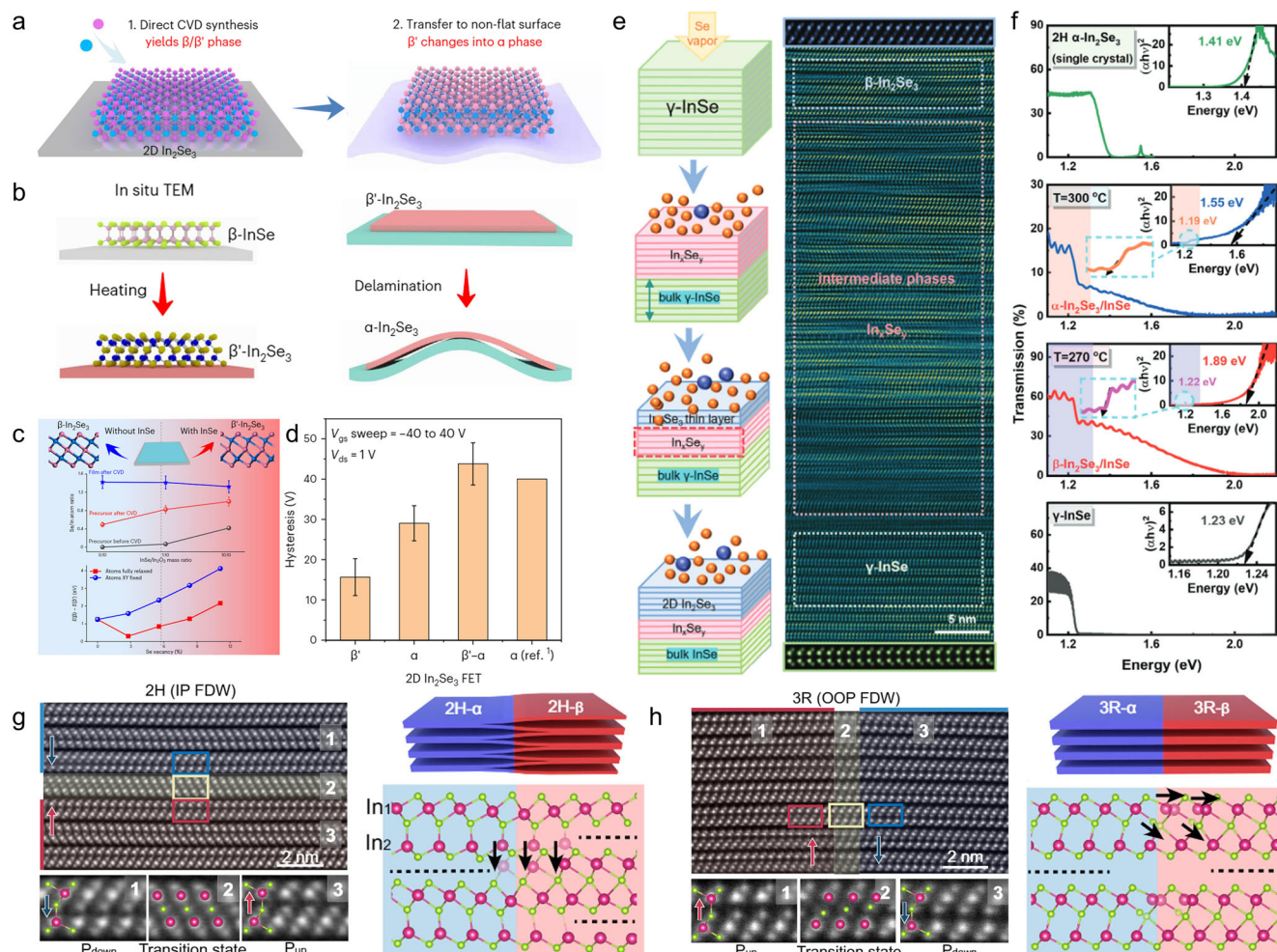


Fig. 4 | Sliding ferroelectricity of noncentrosymmetric crystal phases. **a** Schematic of controlled synthesis for three In_2Se_3 phases (β , β' , α). **b** In situ heating TEM schematic showing β -InSe transformation to β' - In_2Se_3 . **c** Mechanism of phase control through Se/In ratio modulation. **d** Comparison of memory window sizes in devices fabricated from different phases. Reproduced with permission⁷. Copyright 2023, Springer Nature. **e** Transformation process from InSe to In_2Se_3 with

corresponding STEM images. **f** Raman spectra of products grown at 270 °C, 300 °C, and 600 °C selenization temperatures. Reproduced with permission⁶¹. Copyright 2024, John Wiley & Sons, Ltd. **g** Atomic-scale characterization of ferroelectric domain structure and domain wall motion in 2H α - In_2Se_3 . **h** Atomic-scale characterization of ferroelectric domain structure and domain wall motion in 3R α - In_2Se_3 . Reproduced with permission⁶². Copyright 2024, Springer Nature.

alternative approaches such as mechanical exfoliation followed by artificial restacking. For example, Lin et al. developed a “flip-and-reflection transfer” technique to achieve precise zigzag-edge alignment between hBN and graphene, realizing 100% success rate in creating oblique-stacked trilayer hBN/graphene/hBN heterostructures (Fig. 5d)³⁹. Control experiments with non-oblique configurations, including two anti-parallel aligned hBN flakes (flip without reflection) but with graphene aligned and completely misaligned hBN/graphene/hBN stack, showed complete disappearance of ferroelectricity, conclusively demonstrating that oblique stacking is essential for breaking both inversion and mirror symmetries to generate net OOP polarization. Moreover, they claimed that the hBN-encapsulated heterostructures exhibited robust room-temperature ferroelectricity not only for monolayer graphene but also for bilayer and trilayer graphene (Fig. 5e), with all three systems showing clearly switchable polarization responses. Specially, monolayer graphene in hBN-encapsulated devices exhibiting a record-high room-temperature polarization of $1.76 \mu\text{C}/\text{cm}^2$ among sliding ferroelectrics (Fig. 5f) and excellent endurance ($>50,000$ switching cycles). Moreover, the system demonstrates a unique layer-dependent odd-even effect, showing non-monotonic thickness dependence of polarization ($0.81 \mu\text{C}/\text{cm}^2$ for bilayer and $1.22 \mu\text{C}/\text{cm}^2$ for trilayer graphene) that originates from cooperative interlayer sliding between graphene and hBN layers (Fig. 5g, h).

Twist Stackings or Moiré Superlattices

The creation of moiré superlattices through controlled twist stacking has emerged as a transformative approach to induce sliding ferroelectricity in vdW materials that are intrinsically nonpolar in their natural bilayer form. This method operates through a fundamental symmetry-breaking mechanism: when two crystal layers are rotationally misaligned at small angles, the resulting moiré pattern generates periodic domains with alternating AB and BA stacking registries. These distinct stacking configurations produce opposing OOP dipoles due to localized interlayer charge redistribution, while the domain walls between them serve as natural channels for polarization switching *via* interlayer sliding.

The moiré approach offers several unique advantages over conventional ferroelectric design strategies, including universal applicability, tunable ferroelectricity, and multi-functionality. While challenges remain in achieving uniform moiré patterns over large areas and improving operational temperature ranges, twist engineering represents a paradigm shift in ferroelectric materials design by demonstrating that sophisticated control over atomic registries alone can transform ordinary bilayers into functional ferroelectrics, opening new possibilities for novel electronic, optoelectronic, and quantum devices.

As a representative sliding ferroelectric system, hBN exhibits unique polarization control through twist engineering. While monolayer hBN

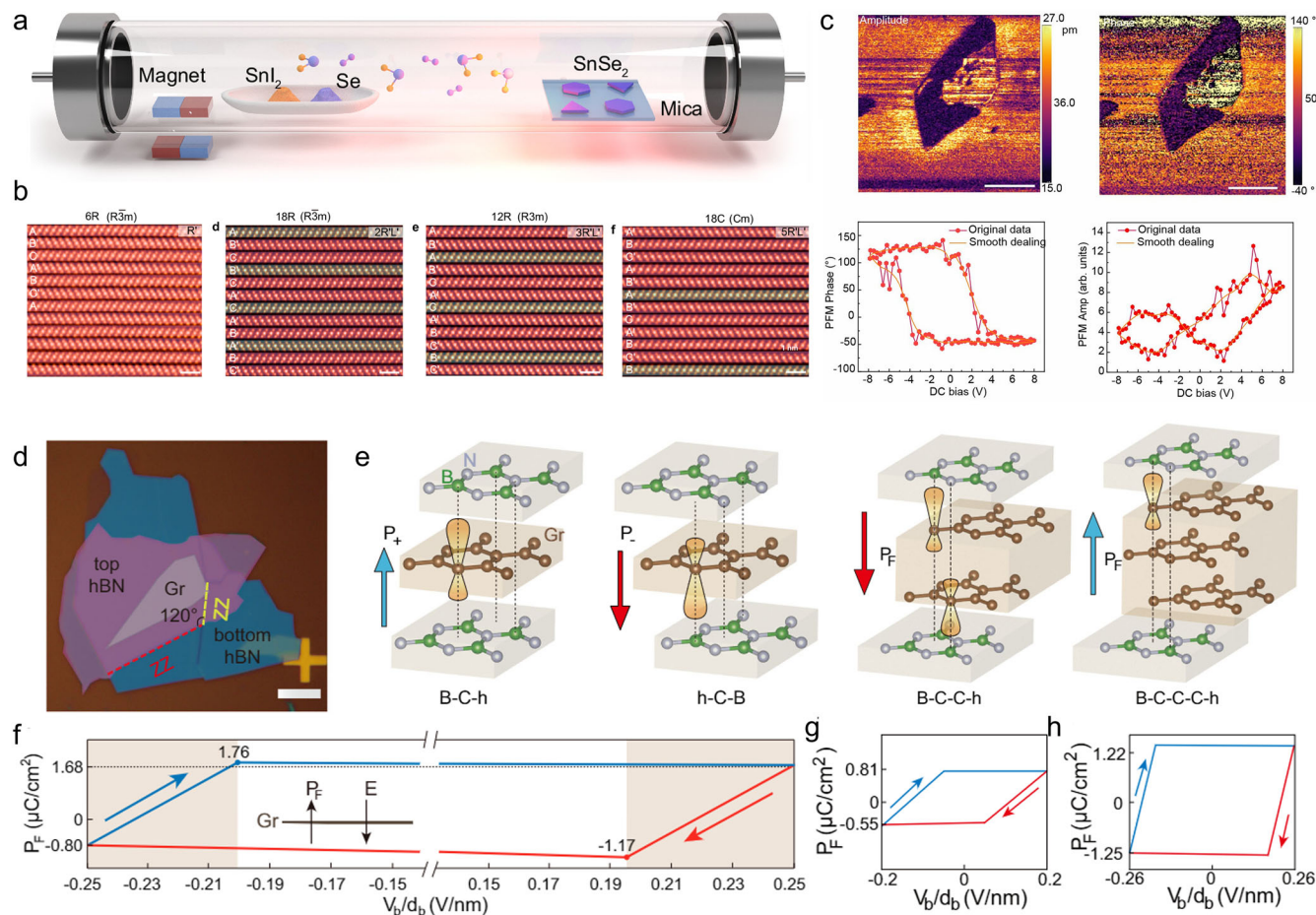


Fig. 5 | Asymmetric stacking strategy for inducing sliding ferroelectricity. **a** The schematic illustration of the substrate-guided CVD growth approach. **b** The experimental atomic structures of 6R, 12R, 18R, and 18C stacking configurations. **c** Ferroelectricity of AB^2 -stacked SnSe_2 . Reproduced with permission⁶⁴. Copyright 2024, Springer Nature. **d** Optical microscope image of hBN-encapsulated graphene.

e Schematic diagram of the crystal structure and ferroelectric polarization direction in monolayer, bilayer, and trilayer graphene intercalation. **f–h** Ferroelectric polarization intensity of graphene with different layer numbers. Reproduced with permission³⁹. Copyright 2025, Springer Nature.

maintains a graphene-like hexagonal lattice with alternating boron (B) and nitrogen (N) atoms, its natural AA' stacked bilayer configuration preserves inversion symmetry (Fig. 6a), resulting in a nonpolar state²⁶. Artificial conversion to AA stacking breaks this symmetry (Fig. 6b) to create polar character, though this configuration remains metastable⁶⁵. Cross-sectional analysis of native AA'-hBN (Fig. 6c) reveals perfect B/N registry symmetry, whereas deliberately stacked AB and BA configurations produce diagonally aligned polarization units that dramatically alter ferroelectric properties. Registration mapping of engineered stacks with 0.5° twist angle (Fig. 6d) shows that lattice relaxation reorganizes moiré patterns into distinct domains: triangular AB/BA ferroelectric regions with opposite polarization directions (due to their mirror-symmetric relationship) separated by AA saddle points and domain walls²⁶. This AB/BA duality in twisted hBN creates a natural platform for polarization switching, where interlayer sliding transitions between these complementary states enable nonvolatile dipole reversal without requiring ionic displacement.

In 2022, Wang et al. achieved a breakthrough by employing twist-stacking technology to create precisely controlled 0.25° -twisted bilayer MoS_2 ²⁷. This ultra-precise angular manipulation successfully broke the local inversion symmetry in bilayer MoS_2 , thereby inducing switchable ferroelectric domains—a phenomenon subsequently verified across multiple transition metal dichalcogenides including WSe_2 , MoSe_2 , and WS_2 .

In 2024, Lee et al. conducted a systematic investigation of twist-angle dependence on ferroelectricity in bilayer WSe_2 , revealing a critical angular

threshold for polarization switching⁶⁶. Their work demonstrated robust room-temperature ferroelectricity at small twist angles (0° – 3°), with complete suppression of ferroelectric signals occurring at $\geq 4^\circ$ misalignment. Most remarkably, at the optimized 3° twist angle, the WSe_2 bilayer exhibited coexisting ferroelectric and ferromagnetic orders—a rare multiferroic combination that opens new avenues for designing multifunctional quantum devices. The observed sharp transition at 4° suggests a delicate balance between interlayer coupling and domain size in stabilizing moiré ferroelectricity. These findings underscore how moiré superlattice engineering is advancing fundamental materials physics while enabling unprecedented control over quantum phenomena for next-generation electronics.

Non-Stoichiometric Compositions

Doping and intercalation strategies have been widely employed to modulate and optimize the ferroelectric properties of conventional intrinsic ferroelectric materials, such as CuInP_2S_6 ⁶⁷. Inspired by this, non-stoichiometric compositions has represented a unique pathway to engineer sliding ferroelectricity by intentionally introducing controlled atomic vacancies, dopants, intercalation, or compositional gradients in layered materials. Unlike conventional approaches that rely on precise stacking sequences or moiré engineering, non-stoichiometry induces ferroelectricity through local symmetry breaking and interfacial charge imbalance between layers. This approach significantly expands the materials design space beyond ideal stoichiometric crystals, bridging the gap between traditional displacive ferroelectrics and emergent sliding mechanisms.

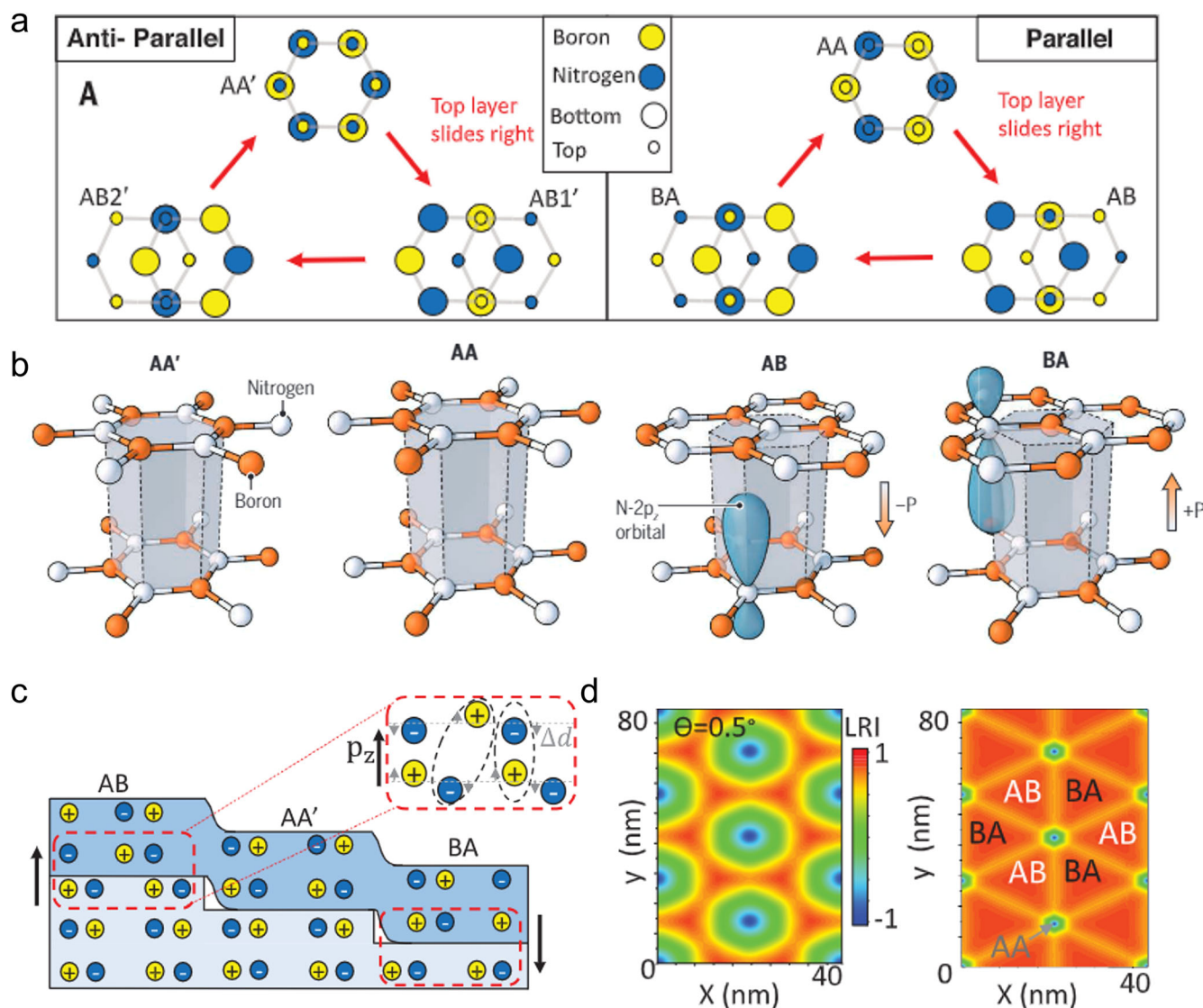


Fig. 6 | Sliding ferroelectricity of twisted stackings and Moiré superlattices. **a** Top views of antiparallel and parallel stacking configurations, showing AB and BA structures formed by lateral sliding of the top layer. Reproduced with permission²⁶. Copyright 2021, American Association for the Advancement of Science. **b** Side views of AA' (antiparallel), AA (parallel), and polar AB/BA stacking configurations. Reproduced with permission⁶⁵. Copyright 2021, American Association for the Advancement of Science.

Advancement of Science. **c** Cross-sectional atomic arrangements of naturally grown AA' and post-sliding AB/BA configurations, where +/- signs represent B/N atoms. While AA' remains fully symmetric, AB and BA exhibit diagonal charge ordering that generates polarization. **d** Initial registration index (left) and post-relaxation mapping (right) of 0.5° twisted bilayer hBN. Reproduced with permission²⁶. Copyright 2021, American Association for the Advancement of Science.

Sui et al. demonstrated that yttrium (Y) doping could significantly enhance the ferroelectric properties of γ -InSe through controlled microstructure engineering²⁹. The Y incorporation improved crystal quality by eliminating stacking faults while inducing two critical structural modifications, in-plane compressive distortion and interlayer pre-sliding configurations, which collectively amplified both OOP and in-plane spontaneous polarization. The broken inversion symmetry was conclusively verified by enhanced SHG signals (Fig. 7a), while PFM measurements demonstrated characteristic ferroelectric switching behavior through both amplitude butterfly loops and phase hysteresis (Fig. 7b, c). Remarkably, 50-nm-thick InSe:Y nanosheets exhibited an effective piezoelectric coefficient (d_{33}) of ~ 7.5 pm/V, which is an order of magnitude enhancement over undoped γ -InSe, highlighting the dramatic impact of Y-mediated structural tailoring on electromechanical performance.

In 2023, Gong et al. demonstrated the widespread existence of piezoelectric/ferroelectric behavior in a series of non-stoichiometric TMDs⁶⁸. Their study on $\text{Fe}_{1+\alpha}\text{Te}_2$ revealed that varying stoichiometric ratios (α) could effectively modulate both OOP and in-plane ferroelectricity (Fig.

7d, e), enabling applications in multi-level memristors. The observed effects originated from two synergistic mechanisms: (i) Frankel defects induced by intercalated metal atoms and (ii) sliding-governed polarization switching at vdW gaps.

This paradigm was further advanced in 2024 by Meng et al., who synthesized non-stoichiometric ternary TMDs ($\text{AgCr}_{1-x}\text{S}_2$) via CVD and confirmed their room-temperature ferroelectricity⁶⁹. SHG measurements verified the broken inversion symmetry, while PFM data demonstrated voltage-enhanced switching: increasing the applied bias from 2 V to 5 V enlarged the hysteresis window and sharpened the butterfly curves. Crucially, Cr vacancies and interlayer atomic displacements were identified as the primary drivers of polarization, with devices exploiting both in-plane and OOP polarizations for multifunctional operation.

Heterostructures

Heterostructures create sliding ferroelectricity through interfacial charge asymmetry and stacking-dependent polarization at vdW interfaces, where dissimilar layered materials break inversion symmetry via controlled

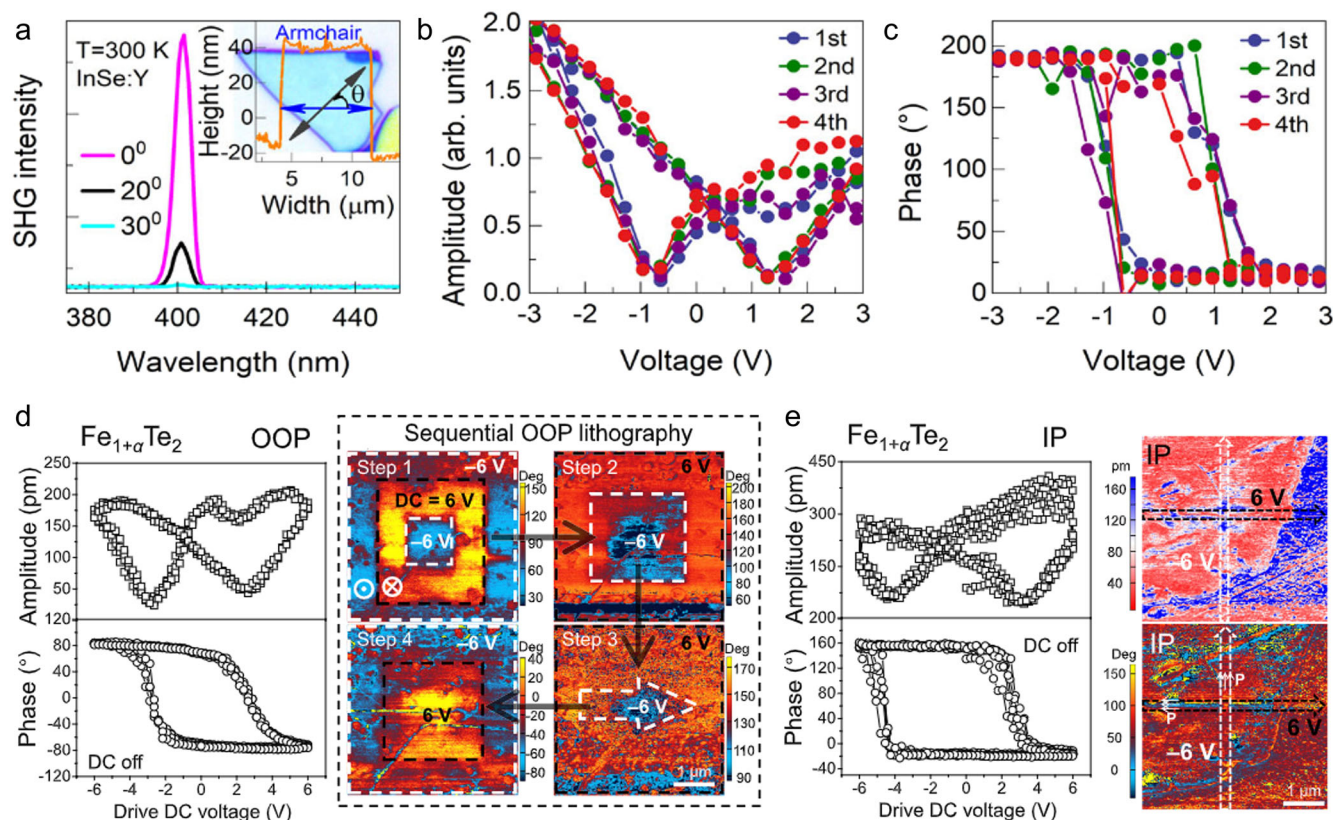


Fig. 7 | Sliding ferroelectricity of non-stoichiometric compositions. **a** Angle-resolved SHG signals of InSe:Y. **b** Amplitude butterfly loops of InSe:Y. **c** Phase hysteresis curves of InSe:Y. Reproduced with permission²⁹. Copyright 2023, Springer Nature.

d OOP ferroelectricity in non-stoichiometric $\text{Fe}_{1+\alpha}\text{Te}_2$. **e** In-plane ferroelectricity in $\text{Fe}_{1+\alpha}\text{Te}_2$. Reproduced with permission⁶⁸. Copyright 2023, Springer Nature.

interlayer registry. Unlike homogeneous bilayers, heterostructures (e.g., MoS_2/WS_2 , graphene/hBN) leverage intrinsic property mismatches—such as work function differences, band alignment, or ionic polarities—to generate robust OOP dipoles that can be switched through interlayer sliding.

Complementing the earlier discussion of graphene/hBN heterostructures, the MoS_2/WS_2 system developed by Rogée et al. reveals distinct advantages in sliding ferroelectric engineering⁴³. Unlike the graphene/hBN system where polarization arises primarily from interfacial charge transfer, the MoS_2/WS_2 heterobilayer achieves its remarkable properties through two synergistic mechanisms: (i) intrinsic symmetry breaking from the lattice-mismatched interface and (ii) interlayer sliding-enabled polarization switching, all without requiring artificial twist angles. Their precisely engineered heterointerface breaks spatial inversion symmetry through interlayer charge redistribution, generating an exceptional OOP polarization ~6 times stronger than monolayer $\alpha\text{-In}_2\text{Se}_3$, a record value among 2D ferroelectric systems at the time of publication.

Microscopic Mechanisms of Sliding Ferroelectricity

The microscopic origin of sliding ferroelectricity fundamentally differs from conventional displacive ferroelectricity, relying instead on interlayer charge asymmetry created by specific atomic registries in vdW layered materials. This phenomenon was first demonstrated in WTe_2 by Fei et al., where bilayer and thicker samples exhibited clear ferroelectric bistability under vertical electric fields while monolayers remained nonpolar, conclusively establishing the role of interlayer sliding rather than ionic displacements²¹. Subsequent first-principles calculations by Yang et al. revealed that polarization switching in WTe_2 occurs through a 0.72 Å lateral shift between mirror-symmetric states, with Hirshfeld charge analysis showing uncompensated interlayer charge transfer ($\Delta q \approx 0.07 e$) as the source of OOP dipoles⁵⁸.

Recent advances have further elucidated the unconventional switching dynamics in sliding ferroelectrics: (i) mechanical bending has been shown to create topological domain walls that reverse polarization via kink formation in hBN and 3R- MoS_2 , distinct from flexoelectric effects⁷⁰. (ii) for BN bilayers, inclined electric fields are found to enable deterministic switching with reduced critical field strength by leveraging in-plane polarization in transition states, suggesting a universal approach for D3h-symmetric monolayers⁷¹. (iii) the unique role of off-diagonal Born effective charges in hBN explains its avalanche-like switching dynamics and wriggling domain wall motion under symmetry-breaking perturbations⁷². These studies collectively reveal that sliding ferroelectric switching is governed by fundamentally different rules compared to conventional ferroelectrics.

Parallel insights emerged from studies of 3R- MoS_2 , where AB/BA stacking configurations (Fig. 8a, b) generate opposing polarizations through interfacial charge transfer, with multilayer systems exhibiting complex switching dynamics due to sequential layer motion and transient antiparallel dipole states during polarization reversal^{54,73,74}. In 2022, Meng et al. made significant advances in understanding the polarization switching dynamics of multilayer 3R-phase MoS_2 ⁷³. Their work revealed an anomalous polarization state arising from the sequential sliding of individual atomic layers - when one layer completes its sliding motion, it creates an intermediate state with antiparallel OOP dipole moments, as illustrated in Fig. 8c. This discovery highlighted the complex, layer-by-layer nature of polarization reversal in multilayer sliding ferroelectrics.

More recently, employed optical reflectance spectroscopy to systematically map the polarization switching pathways in trilayer 3R- MoS_2 ⁷⁴. Their study demonstrated that the sequential depinning of domain walls at different interfaces plays a critical role in determining the overall polarization reversal process, providing new insights into the microscopic mechanisms governing switching dynamics in these materials.

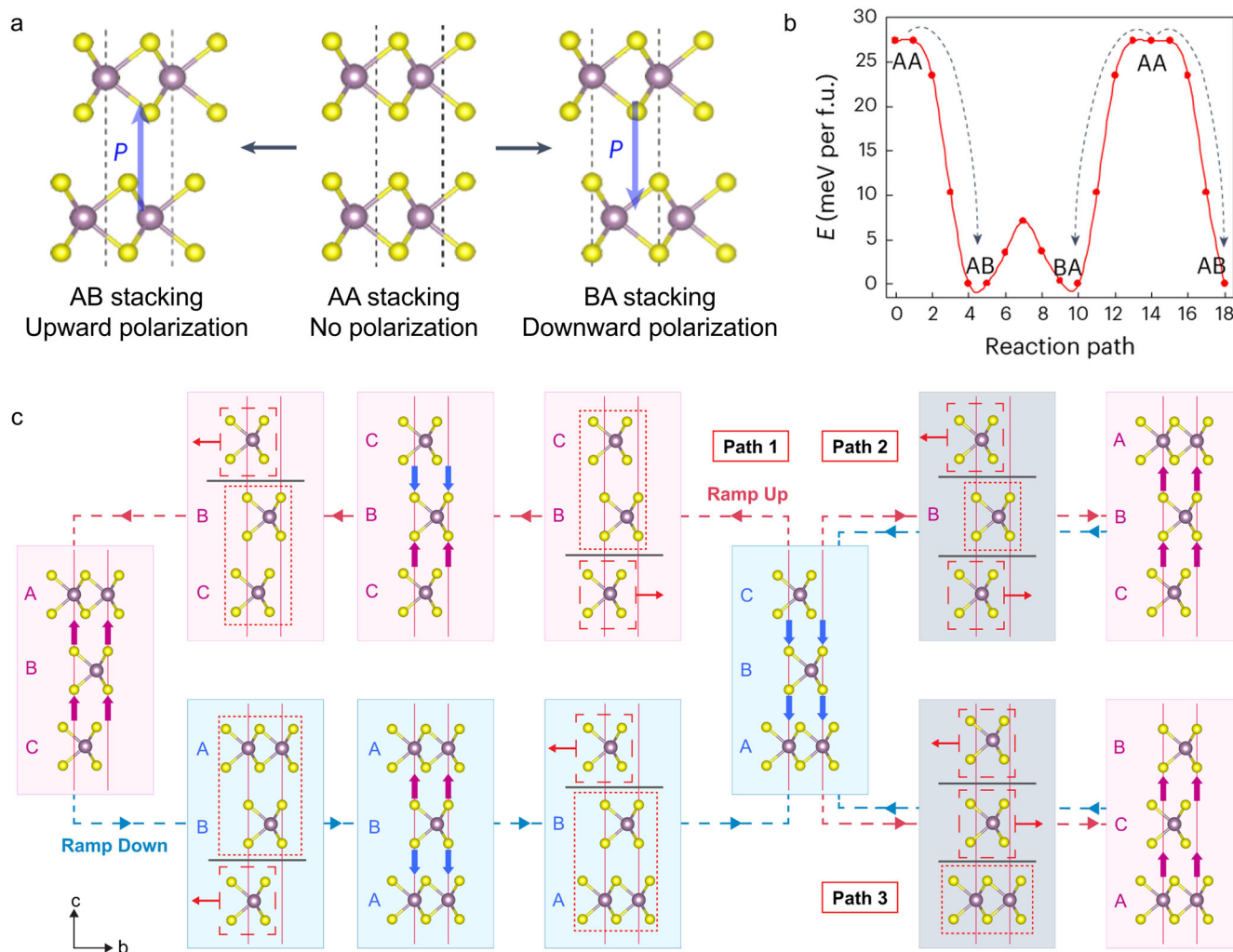


Fig. 8 | Microscopic mechanisms of sliding ferroelectricity. **a** Mechanism of ferroelectric polarization generation in bilayer 3R-phase MoS₂. **b** Thermodynamic energy barriers for different stacking configurations in bilayer 3R-MoS₂, among those AB and BA are two most stable stackings. Reproduced with permission⁵⁴.

Copyright 2024, Springer Nature. **c** Schematic of layer-by-layer sliding mechanism for ferroelectric polarization in multilayer 3R-MoS₂. Reproduced with permission⁷³. Copyright 2022, Springer Nature.

These findings collectively highlight how sliding ferroelectricity universally arises from correlated electron-phonon interactions at vdW interfaces, while manifesting material-specific characteristics - from WTe₂'s layer-number threshold to 3R-MoS₂'s metastable intermediate configurations - that can be tailored through precise stacking control and thickness engineering. The existence of such diverse yet unified mechanisms across different material systems underscores sliding ferroelectricity as a distinct class of ferroelectric phenomena enabled by the unique physics of two-dimensional heterointerfaces.

Device Applications of Sliding Ferroelectric Materials

Current research on device applications based on sliding ferroelectric materials remains at an early developmental stage, yet the preliminary breakthroughs already demonstrate their transformative potential for next-generation electronics. Unlike conventional ferroelectric devices limited by interfacial defects and scaling challenges, sliding ferroelectrics offer three inherent advantages for practical implementation: (i) atomic-scale thickness compatibility with vdW integration, (ii) mechanical flexibility enabling bendable electronics, and (iii) fatigue-resistant switching through defect-tolerant sliding mechanisms.

In 2024, Li et al. achieved a significant breakthrough by fabricating rewritable ferroelectric transistor arrays based on bilayer 3R-phase MoS₂⁴⁴. Leveraging the non-volatile memory characteristics and exceptional

mechanical stability of this sliding ferroelectric material, they successfully demonstrated flexible device integration that maintained robust performance through over 1000 bending cycles (Fig. 9a–d), showcasing unprecedented fatigue resistance for atomic-scale memory elements.

Ferroelectric tunnel junctions (FTJs), which utilize polarization-dependent tunneling resistance for non-volatile memory operation, represent another promising application platform for sliding ferroelectrics. The atomically thin nature of these materials allows for ultra-narrow tunnel barriers, while the reversible OOP polarization enables efficient modulation of tunneling currents⁷⁵. Recent theoretical studies suggest that vdW FTJs based on sliding ferroelectrics could achieve large tunneling electro-resistance ratios⁷⁶, making them attractive candidates for low-power, high-density memory devices.

Parallel advances have been made in optoelectronic applications, as exemplified by recent work on ϵ -InSe-based sliding ferroelectric devices³⁰. This system exhibits unique layer-dependent ferroelectricity, where OOP polarization emerges only in even-numbered layers, enabling both non-volatile memory operation (with on-off ratios $\approx 10^4$) and ultrafast optoelectronic functionality. Remarkably, the ferroelectric polarization in ϵ -InSe induces an ultrafast bulk photovoltaic response (~ 3 ps) in the near-infrared regime (Fig. 9e–h), achieved through polarization-modulated photo-generated carrier dynamics. These findings, supported by first-principles calculations and experimental characterizations, establish sliding

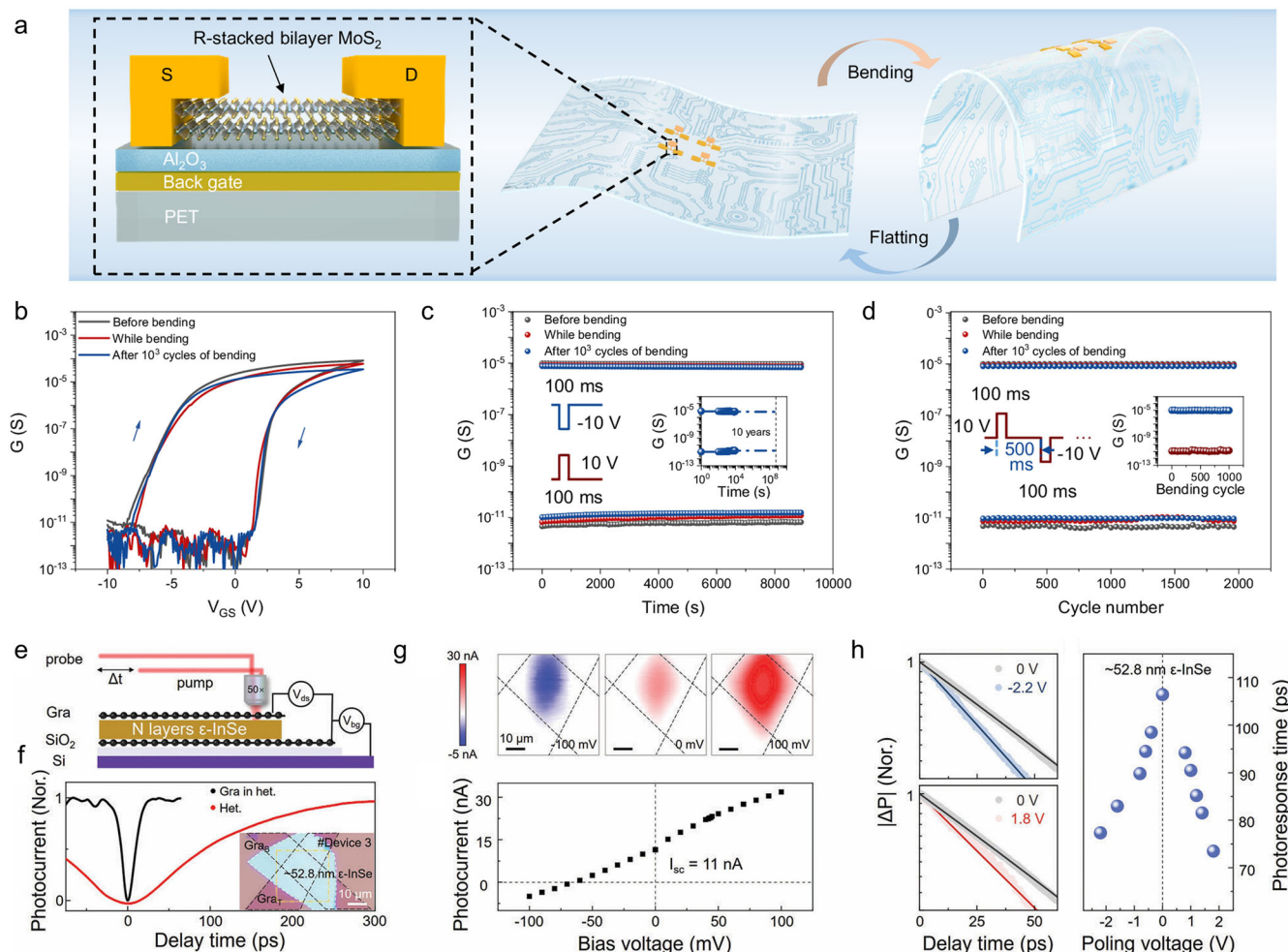


Fig. 9 | Device applications of sliding ferroelectric materials. **a** The schematic diagram of the flexible devices based on 3R-phase MoS₂. **b–d** Transfer, retention, and endurance characteristics of flexible devices after 1000 consecutive cycles of bending test. Reproduced with permission⁴⁴. Copyright 2024, Springer Nature. **e** Graphene/ε-InSe/graphene device configuration and time-resolved photocurrent

measurement setup. **f** Time-resolved photocurrent signal of 52.8 nm ε-InSe. **g** Photocurrent maps and photocurrent-voltage characteristic of the device. **h** Bias voltage-dependent time-resolved photocurrents. Reproduced with permission³⁰. Copyright 2024, John Wiley & Sons, Ltd.

ferroelectrics as a promising platform for hybrid devices combining reconfigurable memory and high-speed photodetection capabilities.

Furthermore, the dynamic and reversible polarization switching behavior, combined with low energy consumption, makes sliding ferroelectrics ideal candidates for neuromorphic computing, particularly in the design of ferroelectric synaptic devices^{77,78}. In such systems, the ferroelectric polarization can be used to emulate synaptic weight, and the conductance of the channel can be modulated via gate pulses mimicking pre-synaptic spikes. The fatigue-resistant nature of sliding ferroelectrics ensures long-term stability and endurance—critical requirements for artificial neural networks. Recent studies have shown that synaptic transistors made from few-layer sliding ferroelectric materials can accurately mimic the behavior of biological synapses by precisely adjusting their electrical response to voltage pulses^{44,79}. This high precision in simulating learning processes makes them promising candidates for building efficient AI systems.

Collectively, these developments highlight the versatility and scalability of sliding ferroelectric materials across a wide range of electronic and optoelectronic platforms, extending well beyond traditional ferroelectric functionalities.

Challenges and Perspectives

The field of 2D sliding ferroelectric materials has witnessed remarkable theoretical and experimental progress in recent years, yet fundamental

challenges persist across material discovery, mechanistic understanding, and functional control. While recent theoretical advances, particularly the general BSF theory⁶³, have established symmetry-based design rules for sliding-induced polarization, translating these principles into experimental systems faces three key hurdles: (i) material limitations: the BSF theory predicts numerous candidate systems (e.g., centrosymmetric monolayers like CrI₃), yet synthesis challenges (e.g., controlled stacking, interfacial defects) restrict experimental validation to a few prototypes (e.g., 3R-MoS₂, twisted hBN). (ii) dynamic complexity: even with known symmetry constraints, the real-time kinetics of polarization switching—especially in multilayers with competing domain walls or rotatable subunits (e.g., CCC crystals)—remain poorly quantified. (iii) property trade-offs: designed sliding ferroelectrics often face conflicting requirements (e.g., strong polarization vs. low switching barriers, stability vs. tunability), necessitating beyond-symmetry considerations like strain engineering or doping. Mechanistically, the interplay between interlayer charge transfer and lattice dynamics during polarization switching remains incompletely understood, particularly in multilayer systems where intermediate states and domain wall dynamics complicate the switching pathways.

In addition, the field of sliding ferroelectrics faces several fundamental challenges that represent truly unresolved frontiers in condensed matter physics and materials science. Beyond current symmetry-based theories, the field must address the many-body problem in switching dynamics, where

electronic correlation effects create >100 meV discrepancies between predicted and observed switching barriers, while defect-mediated nucleation dominates real-world behavior. For twist-angle engineered systems, an additional fabrication challenge emerges: while state-of-the-art transfer stages can achieve 0.1° stacking precision through mechanical exfoliation/restacking⁸⁰, this approach fundamentally limits scalability. Although CVD growth has recently produced large-area twisted bilayers^{81–84}, achieving the crucial 0.1°–5° twist angles—where most sliding ferroelectric phenomena emerge—remains exceptionally challenging due to thermodynamic instabilities during growth. Therefore, developing more reliable techniques to overcome thermodynamic instabilities and enable large-scale preparation of materials with small-angle twist is also a promising research direction. Besides, several reports about the large-scale preparation of 3R-phase bilayer TMDs have emerged⁸⁵. These scalable fabrication techniques pave the way toward the realization of large-scale sliding ferroelectrics and are also considered to be a highly promising research topic. Furthermore, a critical characterization gap persists, with no existing technique achieving the required trifecta of <10 fs temporal resolution, Ångström-scale spatial precision, and meV energy resolution to fully probe switching events. The intrinsic scalability dilemma presents another fundamental hurdle, as material systems cannot simultaneously achieve both large polarization (>50 $\mu\text{C}/\text{cm}^2$) and low switching fields (<0.1 MV/cm) due to competing requirements for strong versus weak interlayer coupling.

Looking ahead, three key directions emerge as particularly promising for advancing the field of sliding ferroelectrics: (i) theory-guided discovery that combines the BSF framework with high-throughput ab initio screening to identify systems with both synthesis-compatible stacking configurations and robust switching metrics; (ii) operando characterization utilizing cutting-edge techniques like ultrafast TEM or X-ray free-electron lasers to resolve switching dynamics at the Ångström-femtosecond scale, thereby bridging the gap between theoretical predictions and experimental observations; and (iii) development of hybrid architectures that exploit sliding ferroelectricity in heterostructures—such as those coupled with 2D magnets or superconductors—to unlock novel emergent phenomena including electrically controlled topological phases. These interconnected approaches will collectively drive the field toward both fundamental understanding and practical applications.

From a broader perspective, the unique characteristics of sliding ferroelectrics, such as their thickness-dependent polarization, mechanical flexibility, and potential for ultralow-power switching, position them as a transformative materials platform that could bridge the gap between conventional ferroelectrics and emerging quantum materials. Realizing this potential will require coordinated efforts to establish synthesis-structure-property relationships, develop standardized characterization protocols, and explore unconventional device architectures that leverage their distinctive physics. As these efforts progress, sliding ferroelectrics may not only address current technological challenges in non-volatile memory and energy-efficient electronics, but also open new frontiers in the control of quantum phases and non-equilibrium material states.

Data availability

No datasets were generated or analysed during the current study.

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Author contributions

D.G. and H.M. conceived and supervised the project. Q.Z. wrote the original draft, A.F., Y.W., F.W., and L.L. reviewed the draft. All authors discussed the results and commented on the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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